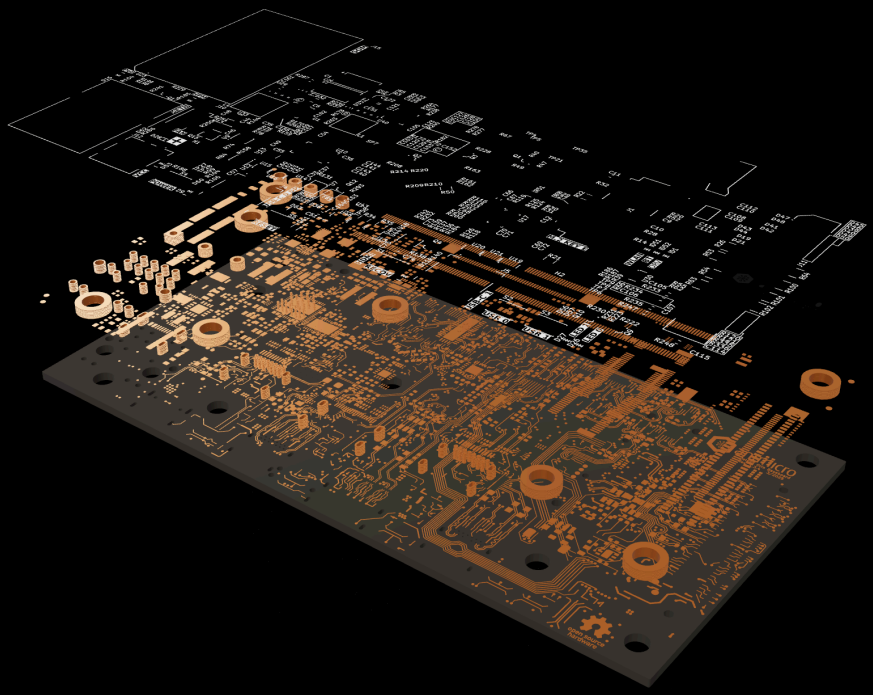
Rauchwerk Research

FIELD REPORT · NOVEMBER 2026
LITERATURE REVIEW · EVAL · FORECAST



PCB GOLD-MASTER EVAL

Routed is not the same as right

Self-improving machine learning for high-frequency PCB placement and routing: what the literature proves, which open-source tools can carry it, a new eval that scores machine routing against eight fabricated Antmicro boards, and what an outcome costs in dollars and hours.

DAVE RAUCHWERK · RAUCHWERK RESEARCH

1 THE EVIDENCE
2 THE TOOLS
3 THE EVAL
4 THE FORECAST

AT A GLANCE · THE ANSWER, IN ONE LINE

Machines can connect a board. Nothing open yet routes it the way its engineers did, and the eval to prove it did not exist.

The literature has no learned PCB method that optimises impedance, skew or length matching; only vendors claim it. So the work started with the measuring stick: a Harbor eval that strips the copper from eight fabricated, Apache-licensed Antmicro boards and scores machine routing against the human original.

8 gold-master boards, 4–12 copper layers, 20–797 parts, pinned by commit and sha256	**0.27** Freerouting on the M.2 PCIe adapter: 96% routed, 1 of 9 pairs in skew tolerance	**1.00** oracle score: the human copper scores itself perfectly through Harbor
21× CPU-hours RL used over simulated annealing in the only independent chip-placement rerun, for equal or worse results	**$250** to $300 per LLM-evolved placer run (HELIX, ICCAD 2026) for a 4.5–5.6% wirelength gain	**0** open PCB benchmarks with a signal-integrity metric before this one

Five findings

1 Search beats reinforcement learning per dollar, so far.
Evolutionary and tree search won at hour scale in chip placement; RL's headline result has no independent reproduction.

2 Completion hides the high-frequency failure.
The open router finishes 72–96% of simpler boards, stalls at 18% on a BGA board, and misses the skew, length and width targets everywhere.

3 The evaluator is the bottleneck, not the optimiser.
Proxy metrics barely predict final quality (Kendall τ 0.05–0.40); full-wave checks cost minutes per structure.

4 Compute is cheap; wall clock and fidelity are not.
A medium board's routing candidate costs about one cent on CPU. Budgets matter only for RL, LLM evolution and full-wave EM on DDR-class boards.

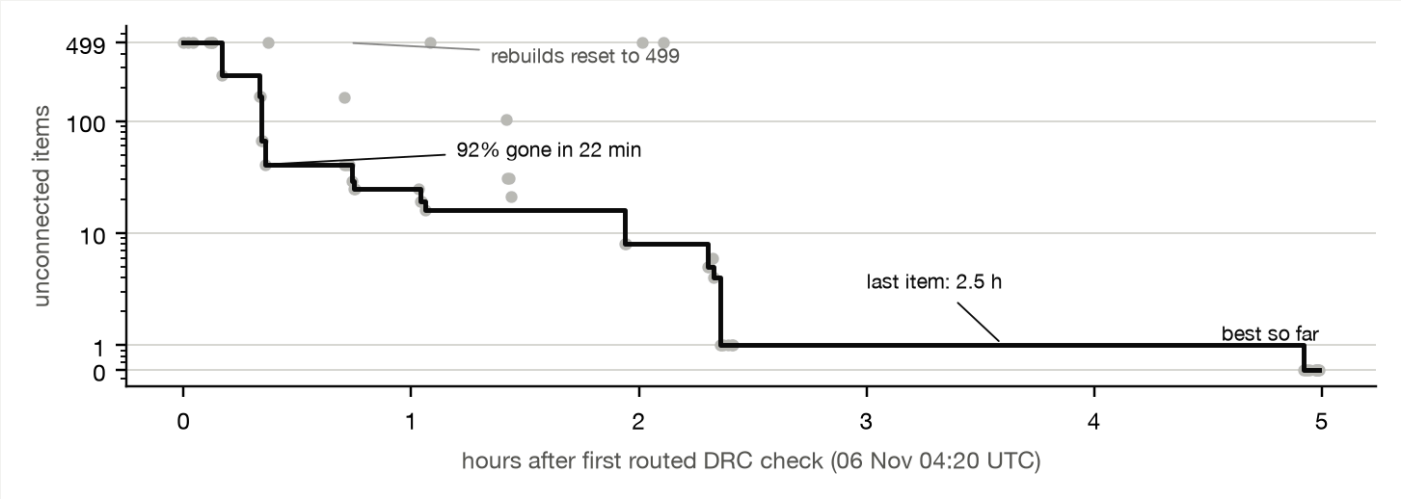
5 Gold masters turn opinion into a number.
Antmicro's length and skew targets exist only in their copper; the eval extracts them and makes every claim testable.

THE QUESTION

Can self-improving machine learning route a high-frequency board as well as the engineers who shipped it?

The claim under review: reinforcement learning, evolutionary algorithms and deep networks that learn to place and route boards for high-frequency constraints, without hand-coded heuristics. Testing it needs three things the field lacks: graded evidence, an open tool stack, and a benchmark with physics in the score.

EXHIBIT 1 THE STARTING POINT: ZINE MACHINE V3, ROUTED BY AN AGENT WITH OPEN TOOLS



Best-so-far unconnected items on the v3 main board, 6 Nov 2026, from the Backplane transcripts. Ninety-two per cent of connections were made in 22 minutes; the last one took 2.5 hours. The 499 ceiling is an artefact: `kicad-cli`'s DRC report stops listing unconnected items at 499, a cap found while building this eval.

WHY A NEW BASE CASE

The project board is a weak test

v3's fastest interface is USB 2.0 and its parallel display bus runs near 30 MHz. It has no DDR, no PCIe and one routed history. A credible test needs boards with real high-speed interfaces, routed by professionals, built and used.

SCOPE

Open source only

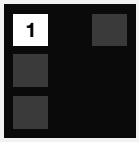
Every tool, solver and dataset here is open source and runs headless in a pinned container. Commercial routers (Quilter, DeepPCB, Allegro X AI) appear only as context, graded as vendor claims.

METHOD

Four parallel workstreams

A graded literature review (30 sources), an inventory of 40 open tools with live repo data, an eval built and validated in Harbor, and a compute forecast with every parameter tagged by source.

CHAPTER 1 OF 4 · WHAT THE LITERATURE PROVES



The evidence favours search, surrogates and honest evaluators over end-to-end RL

EXHIBIT 2 THIRTY SOURCES GRADED; THE ELEVEN THAT DECIDE THE QUESTION

LINE OF WORK	VENUE	GRADE	DECISIVE NUMBER
Chip RL placement (Google CT / AlphaChip)	Nature 2021, 2024	B, disputed	Rerun: 19,068 CPU-h vs 896 for SA; SA best proxy in 6 of 9; no published successful reproduction
Independent reassessment (Cheng, Kahng et al.)	ISPD 2023, TCAD	B	Proxy cost vs routed WL Kendall τ 0.40; vs WNS 0.05
WireMask-BBO (evolutionary)	NeurIPS 2023	A–	73% of gain in first 300 of 2,000 evals; CPU only
EfficientPlace (tree search + RL)	ICML 2024	B	0.5k→1k steps doubles time for 0.3–6.6% more
DREAMPlace (GPU analytical)	DAC 2019	A	>30× faster than RePlAce at equal quality
RL PCB placement (Vassallo, Bajada)	DATE 2024	B	–17 to –21% routed WL vs SA; compute not reported
PCBWorld (RL routing, 679 boards)	KDD wkshp 2026	C	PPO 0.45 vs Freerouting 0.78 clean-pass on harder boards; no SI metric
FunSearch (LLM program search)	Nature 2024	B	~10^6 samples; 4 of 140 runs hit the best result
HELIX (LLM-evolved placer)	ICCAD 2026	B	$250–300, ~120 candidates per run; –4.5% HPWL, –5.6% routed WL
SI surrogates (eye, PDN, EM)	preprints 2021–26	C	R^2 >0.99; 10^3–10^5× faster; NRMSE 10.2→6.6% from 10k→100k samples
Quilter, DeepPCB	vendor	D	Claims of DDR/diff-pair handling; no peer-reviewed results

Grades: A peer-reviewed and independently reproduced · B peer-reviewed · C preprint or non-archival · D vendor claim. Every citation was checked against arXiv, a DOI or the publisher; full bibliography in the companion paper.

Proven

GPU analytical placement is fast and good. Evolutionary and tree search beat pure RL at hour scale. LLM-evolved heuristics give single-digit gains at a known price. Surrogates reproduce SI metrics in narrow parameter spaces.

Not proven, and the gap

“Superhuman” RL placement; any learned PCB router that beats rule-based routers on real boards; any method that optimises high-frequency SI end to end. No PCB benchmark scores signal integrity, so none of it can yet be shown.

FIVE NUMBERS THAT CARRY THE FORECAST IN CHAPTER 4

21×
more CPU-hours for RL than simulated annealing, for equal or worse placements

73%
of WireMask-EA’s gain inside its first 300 of 2,000 evaluations

0.4 vs 2.8%
best-of-120 single shots vs an evolutionary loop, same budget (HELIX)

τ 0.05
rank correlation between proxy cost and timing slack: proxies mislead

10^3–10^5×
surrogate speedup over SI simulation once trained on 10k–100k samples

CHAPTER 2 OF 4 · THE OPEN-SOURCE STACK, SCORED

The open stack is complete except for one piece: fast physics

EXHIBIT 3 OPEN-SOURCE TOOLS SCORED 1–5 ON PERFORMANCE, GPU ACCELERATION, SCRIPTABILITY AND PCB INTEGRATION (REPO DATA 7 NOV 2026)

LAYER	TOOL	LICENCE	PERF	GPU	SCRIPT	PCB	NOTE
Routing	**KiCadRoutingTools**	MIT	4	1	5	5	Rust A*; diff pairs, length/time matching; 0.65 s small boards
Routing	**Freerouting 2.5**	GPL-3	3	1	4	5	Reference autorouter; single-threaded router; varies by run
Routing	**KiCad 10 IPC API / kicad-cli**	GPL-3	3	1	3	5	DRC, export; headless API arrives in KiCad 11
Placement	**DREAMPlace**	BSD-3	5	5	3	1	GPU analytical placer; needs a PCB adapter
EM / SI	**openEMS + gerber2ems**	GPL / Apache	3	3	4	5	Gerbers to S-parameters; GPU engine unreleased
EM / SI	**Palace (AWS)**	Apache-2	5	4	2	1	FEM, multi-GPU; needs a gmsh pipeline
EM / SI	**gprMax / FDTDX**	GPL / MIT	5	5	4	1	Fast GPU FDTD; no PCB ports or import
EM / SI	**scikit-rf, PyBERT**	BSD-3	4	1	5	2	S-parameter cascade; eye and BER
Optimiser	**evosax / EvoTorch**	Apache-2	5	5	5	1	Batched CMA-ES and NES on GPU
Optimiser	**pymoo, Optuna, BoTorch**	Apache / MIT	4	1–4	5	1	Multi-objective and multi-fidelity search
Learning	**PufferLib, SB3, PyG**	MIT	4	4	5	1	PPO and graph encoders
Self-improving	**OpenEvolve, ShinkaEvolve**	Apache-2	3	–	4	1	LLM rewrites router cost functions

Scores from primary sources (READMEs, papers, release notes). GPL tools are called as separate processes so the rest of the stack stays permissive. 28 further tools were reviewed and left out: inactive, chip-only or no PCB path.

Fast loop

KiCadRoutingTools as the inner router, Freerouting as the reference, analytic impedance tables and skew/length checks as the cheap score. Many single-threaded processes, not one big machine.

Physics

openEMS through gerber2ems is the only open path from a PCB to S-parameters. Its GPU engine landed on master in October 2026 with no published speedups: benchmark it before budgeting GPUs.

Self-improvement

evosax or pymoo for routing recipes; OpenEvolve for router code; BoTorch to decide when a candidate earns a full-wave check. RL last, if at all.

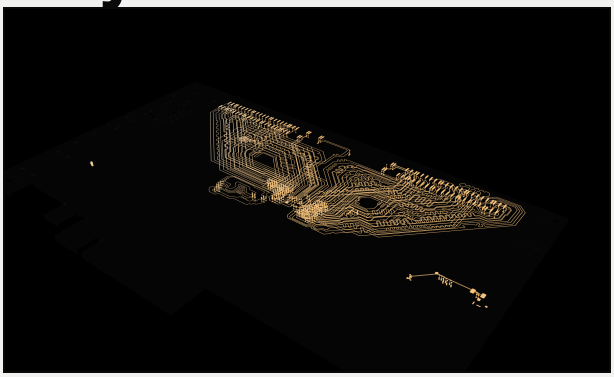
CHAPTER 3 OF 4 · A BENCHMARK WITH PHYSICS IN THE SCORE

Eight boards that were built and used become the answer key

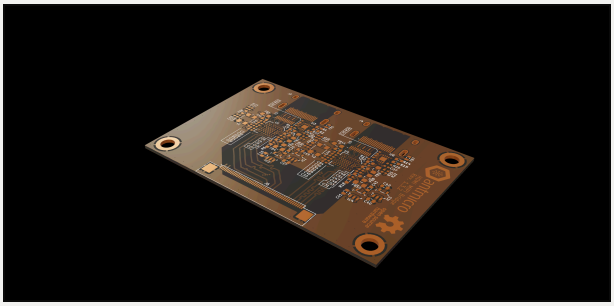
EXHIBIT 4 THE GOLD-MASTER SET: ANTMICRO OPEN HARDWARE, APACHE-2.0

BOARD	TIER	CU	PARTS	PAIR NETS	TO ROUTE	HIGH SPEED	BUILT
SI test board	easy	4	28	2	107	50 Ω test lines	VNA data shipped
M.2 PCIe adapter	easy	4	20	18	93	PCIe x4	not documented
HDMI–MIPI bridge	easy	4	121	36	304	HDMI, CSI-2	photo in repo
GMSL deserializer	medium	4	236	28	413	GMSL2, CSI-2	photo in repo
Jetson Orin baseboard	medium	8	680	208	1,625	PCIe, USB 3, CSI-2, HDMI, GbE	sold (CircuitHub)
RDIMM DDR4 tester	hard	12	691	226	2,021	DDR4 RDIMM	Rowhammer research
SO-DIMM DDR5 tester	hard	12	706	102	1,770	DDR5 SO-DIMM	photo in repo
RDIMM DDR5 tester	hard	12	797	178	2,300	DDR5 RDIMM	VNA vs simulation

“To route”: connections left after the copper is stripped, from KiCad’s connectivity engine. Antmicro’s rule files set widths and gaps but contain no length or skew limits: those targets live only in the human copper, which is why the eval scores against the copper itself.



RDIMM DDR5 tester, data nets only: the length-matching meanders an engineer drew by hand are the answer key.



HDMI-MIPI bridge, four layers, 121 parts: the easy tier.

EXHIBIT 5 HOW A TASK RUNS (HARBOR 0.25 FORMAT, ONE TASK PER BOARD)

1 · Strip

The image build fetches the gold board at a pinned commit, checks its sha256, removes tracks, vias, tuning patterns and fills, and deletes the original in the same layer.

2 · Route

The agent gets the stripped board, KiCad 10.0.7, Freerouting 2.5 and Java 25, and must save a routed board in place without moving a part.

3 · Verify

A separate verifier re-fetches the gold, refills zones and compares, net by net and layer by layer.

4 · Score

`placement × completion² × (0.30 DRC + 0.25 pair skew + 0.15 length groups + 0.15 impedance width + 0.15 efficiency)`

FIRST RESULTS

The open router connects the board and misses the physics.

Freerouting 2.5, run with no agent on the stripped boards, is the floor every method must beat. On boards without dense BGAs it completes 72–96% of connections, then fails exactly where high-frequency design lives: pair skew, length groups and impedance-correct widths. On the BGA-heavy HDMI bridge it stalls at 18%.

EXHIBIT 6 FREEROUTING BASELINE AGAINST THE GOLD MASTERS (BEST OF 18 RUNS; 1.00 = THE HUMAN ORIGINAL)

BOARD	ROUTER TIME	ROUTED	DRC	PAIR SKEW	LENGTH GROUPS	IMP. WIDTH	EFFICIENCY	REWARD
SI test board	54 s	90%	1.00	0.00	1.00	0.00	1.00	**0.48**
M.2 PCIe adapter	112 s	96%	0.18	0.11	0.00	0.43	1.00	**0.27**
HDMI–MIPI bridge	181 s	18%	0.34	0.00	0.00	0.00	1.00	**0.01**
GMSL deserializer	534 s	72%	0.08	0.29	0.00	0.00	1.00	**0.13**

Each column is the share of the human result achieved (DRC: penalty for errors beyond the gold board's own). Runs on 1 CPU core under x86 emulation on an Apple M-series laptop; pass limits 1, 3 and 10, two seeds. Runs vary: on the PCIe adapter, reward ranged 0.21–0.27 across seeds and pass limits. Independent runs overlap the gold copper by about 1% (`gold_copy_fraction` 0.008–0.012). Oracle: 8 of 8 tasks score 1.00 through Harbor.

What works WORKS WELL

The pipeline end to end: pinned inputs, strip, separate verifier, oracle at 1.00, and a gold-copy check that flags any run reproducing the human copper.

What the floor shows REFINE

On the PCIe adapter the router made 96% of connections but only 1 of 9 pairs met the skew tolerance and 43% of high-speed copper used an impedance-correct width.

Next runs COMING SOON

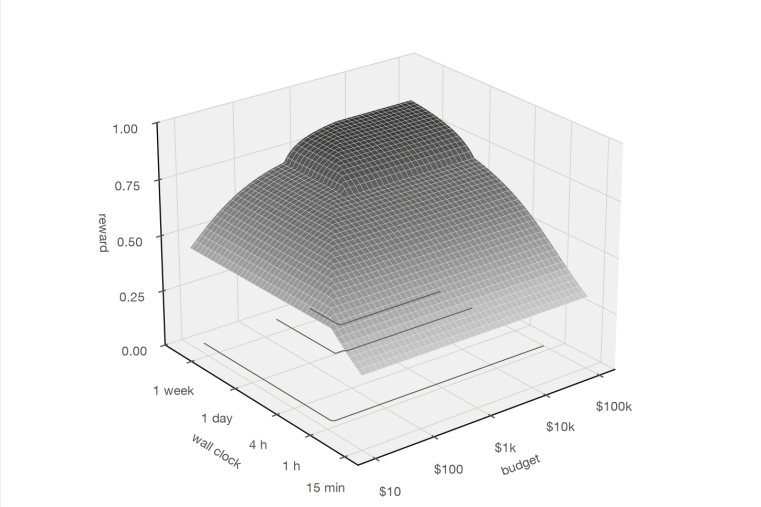
Agent runs through Harbor (Claude Code, Codex, OpenHands), KiCadRoutingTools with length matching, and the hard DDR tier with a full-wave check on sampled nets.

Integrity. Agents have network access, so a run could download the gold board. The verifier reports `gold_copy_fraction`, the share of segments that coincide with the human copper on a 10 µm grid; leaderboards disqualify runs above 0.5. Public boards may also sit in model training data, a contamination risk shared by every open benchmark.

CHAPTER 4 OF 4 · OUTCOME AGAINST DOLLARS AND HOURS

4 Below a few hundred dollars per board, time and the evaluator decide the outcome

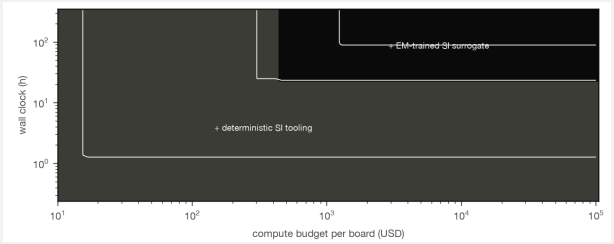
EXHIBIT 7 BEST ACHIEVABLE REWARD, HARD TIER (DDR5-CLASS BOARD)



Envelope over seven strategies; floor contours at 0.3, 0.5, 0.7 and 0.8. The step near $300 is the surrogate's data bill; the ridge near one day is its training time.

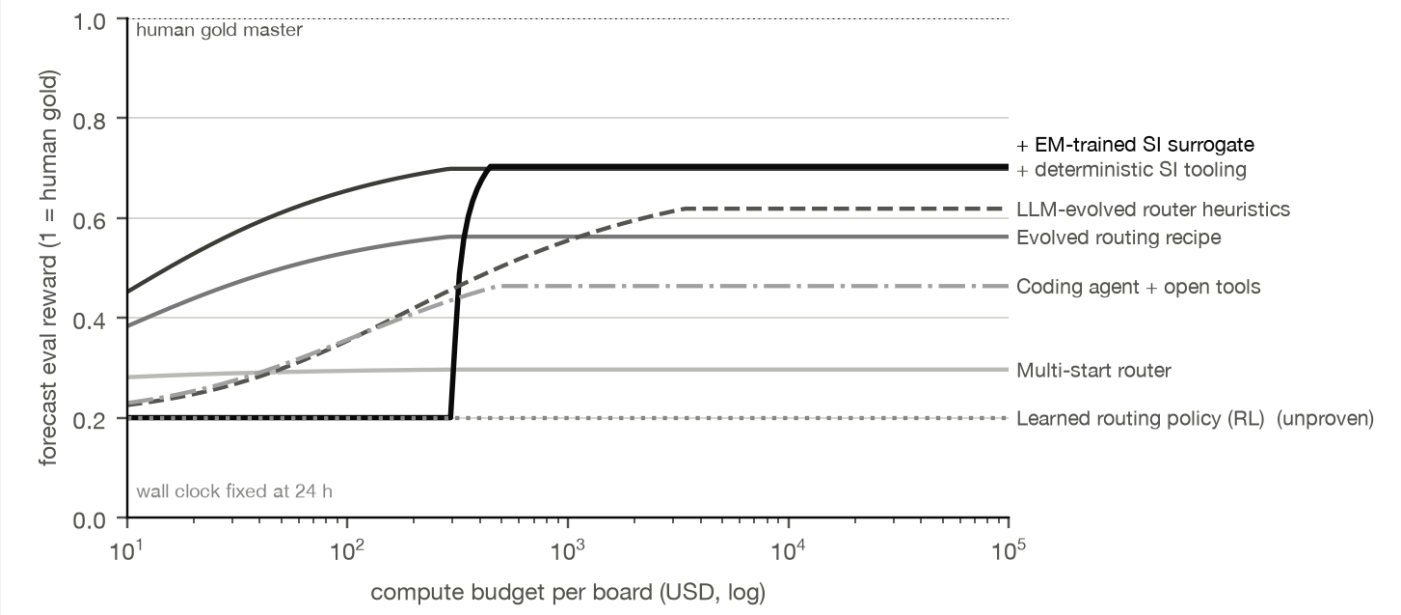
HOW THE FORECAST WORKS

Each strategy has a fixed cost (training data, GPU fit), a cost and duration per candidate, a parallelism limit and a saturating return, $G = g_0 + (g_max - g_0)(1 - (1 + N/n_0)^{-\alpha})$, where N is the number of candidates the budget and the clock both allow. $\alpha = 0.6$ and $n_0 = 60$ reproduce WireMask-EA's diminishing returns. Starting points come from the measured baselines; ceilings for untested strategies are labelled assumptions. Prices: Modal CPU $0.047 per core-hour, A100 $2.50 per hour (7 Nov 2026). All parameters live in `forecast/forecast.toml`; `make forecast` redraws every chart.



Which strategy is best where (hard tier). Black: EM-trained surrogate; dark grey: deterministic SI tooling.

EXHIBIT 8 REWARD AGAINST BUDGET AT 24 HOURS, HARD TIER



BUDGETING RECOMMENDATIONS

Spend on parallel CPU and on the evaluator. Buy GPUs only after the solver earns them.

EXHIBIT 9 FORECAST REWARD BY BUDGET AND WALL CLOCK (MEDIUM · HARD TIER)

BUDGET PER BOARD	4 HOURS	1 DAY	1 WEEK	STRATEGY THAT WINS
$100	0.71 · 0.61	0.78 · 0.66	0.78 · 0.66	+ deterministic SI tooling
$1,000	0.71 · 0.61	0.82 · 0.70	0.84 · 0.79	+ EM-trained SI surrogate
$10,000	0.71 · 0.61	0.82 · 0.70	0.84 · 0.82	+ EM-trained SI surrogate

1 **Cap routing compute at about $100 per board.**
A medium-board candidate costs about one cent of CPU; $100 buys some 9,000 candidates, far past the point where search flattens.

2 **Put the next $1,000 into the evaluator.**
Build the SI tooling and an openEMS-trained surrogate for the hard tier; it is the only strategy whose ceiling rises with spend.

3 **Buy wall clock with cores, not GPUs.**
256 spot cores at about $4 an hour turn a one-week search into an overnight one. Benchmark openEMS's new GPU engine before any GPU spend.

4 **Fund RL only as a research line.**
At about $3,000 and three days of fixed cost, it reaches less than search does today.

EXHIBIT 10 REWARD AGAINST WALL CLOCK AT $1,000, HARD TIER: THE CLOCK, NOT THE BUDGET, SEPARATES THE STRATEGIES

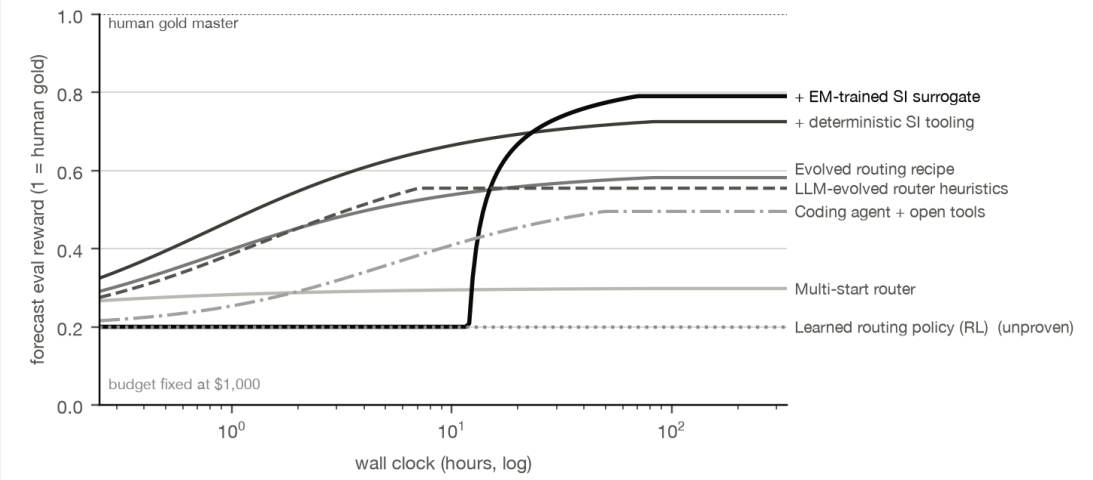


EXHIBIT 11 PHASED PLAN

P0 · this week
Publish the eval; run three coding agents and KiCadRoutingTools through Harbor on the easy and medium tiers. Exit: a public leaderboard.

P1 · 2–3 weeks
Evolved routing recipes (pymoo) with deterministic SI fix-ups. Exit: beat Freerouting by 0.2 on the medium tier.

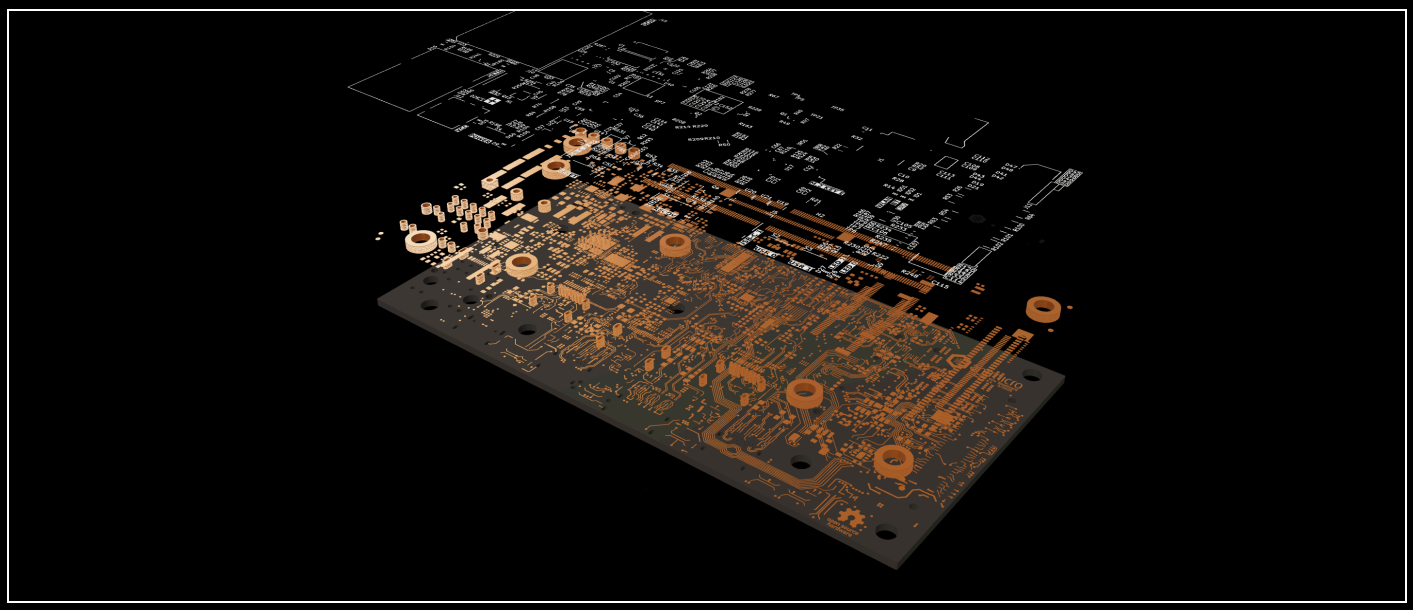
P2 · 1–2 months
gerber2ems sampling and an SI surrogate; full-wave checks on the top 5%. Exit: hard-tier pair and length scores above 0.8.

P3 · quarter
Placement tasks (unplaced boards), LLM-evolved router heuristics, then RL if the data supports it.

THANKS

Measure the physics. Search before you learn. Keep the human copper as the answer key.

Gold-master boards by Antmicro (Apache-2.0), fetched at pinned commits. Built with KiCad 10, Freerouting, Harbor, three.js and matplotlib. Research assisted by AI agents; every figure is reproducible with `make all`.



REPO

github.com/elevenarms/pcb-gold-eval

PROJECT PAGE

pcb-gold-eval.vercel.app

SERIES

Field Report 01 · build loop
Field Report 02 · agentic hardware
Case Study · PLM with agents
Field Report 03 · gold-master eval